

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

**Performance Optimisation of Explainable
Machine Learning Models for Stock Market
Crash Prediction**

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by:

Manya Jain

2210991900

Navdeep Kaur

2210991962

Supervised By:

Dr. Ajay Kumar

Associate Professor

Chitkara University



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA**

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"Performance Optimisation of Explainable Machine Learning Models for Stock Market Crash Prediction"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of our own work carried out under the supervision of **Dr. Ajay Kumar**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Manya Jain

Date: 28 April 2026

(University Roll No. 2210991900)

Navdeep Kaur

(University Roll No. 2210991962)

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Dr. Ajay Kumar

Assistant Professor

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

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I also extend my appreciation to the open-source community behind scikit-learn, XGBoost, SHAP, and imbalanced-learn libraries, whose tools made it possible to implement, experiment, and evaluate the proposed framework efficiently.

Finally, I thank my family and peers for their constant motivation and moral support during this academic journey.

ABSTRACT

The stock market crash is a rare occurrence but can have far-reaching economic impacts. Financial markets are nonlinear dynamic systems with multiple structural breaks and persistent class imbalances, making accurate crash timing prediction extremely challenging. While machine-learning ensembles predict temporal patterns for equities, most operate as black boxes that do not provide the transparency required by regulators or financial institutions.

This report proposes an optimised soft-voting ensemble of Decision Tree, Logistic Regression, Random Forest, and XGBoost classifiers, augmented with SHAP (SHapley Additive exPlanations) for model interpretability. The model was trained on NIFTY 50 stock data (2001-2026) using 6,211 daily trading records, with 25 engineered technical and volatility indicators across five historical crash events, using SMOTE oversampling to address class imbalance.

The ensemble achieved ROC-AUC of 0.627 on the 2020-2023 test set - a 7.5% improvement over the standalone Decision Tree. Top crash precursors identified via SHAP: ATR (14-day), 20-day Volatility, and High-Low Range. Robustness was validated across three distinct market regimes using a 10-quarter rolling window.

Keywords: Stock Market Crash Prediction, Machine Learning, NIFTY 50, Ensemble Methods, SHAP, Explainable AI, SMOTE

1. Introduction

1.1. Background

Financial markets are complex adaptive systems characterised by nonlinear dynamics, extreme events, and structural breaks. Stock market crashes - sudden and severe price declines affecting broad market indices - represent the most disruptive of these extreme events. While crashes are statistically rare, occurring in approximately 10% of all trading days when defined by a 5-day forward return below -3%, their economic consequences are substantial, triggering portfolio losses, liquidity crises, and broader macroeconomic disruptions.

The Indian equity market, represented by the NIFTY 50 index, has experienced five major crash events since 2001: the Dot-com Bust (2001-2002), the Global Financial Crisis (2008-2009), the Euro Debt Crisis (2011), the COVID-19 Pandemic shock (2020), and the Inflation Shock (2022). Each episode demonstrated the inadequacy of reactive risk management approaches and highlighted the demand for proactive, data-driven predictive systems.

1.2. Problem Statement

Despite significant advances in machine learning applications to financial forecasting, two critical gaps persist in the existing literature. First, most high-performing models - particularly deep learning approaches such as LSTMs and Transformers - operate as black boxes, offering no insight into the features driving their predictions. This opacity is incompatible with regulatory frameworks such as the Reserve Bank of India's guidelines on model risk management, which require institutions to explain and justify algorithmic decisions.

Second, ensemble methods for equity crash prediction, while consistently outperforming individual classifiers, have rarely been applied to the NIFTY 50 with rigorous temporal validation across multiple market regimes. The absence of rolling-window validation in prior studies raises questions about whether reported performance reflects genuine predictive power or data leakage from improper train-test splits.

1.3. Objectives

The primary objectives of this research are as follows:

1. To develop a soft-voting ensemble classifier combining Decision Tree, Logistic Regression, Random Forest, and XGBoost for NIFTY 50 crash prediction.

A soft-voting ensemble aggregates the class probability outputs of multiple base classifiers rather than relying on a single model's decision boundary. By combining the interpretability of a Decision Tree, the linear discriminative capacity of Logistic Regression, the variance-reduction strength of Random Forest bagging, and the sequential error-correction of XGBoost gradient boosting, the ensemble is designed to compensate for individual model weaknesses. This diversity of learning strategies is expected to produce more stable and generalisable predictions across varied market conditions than any single classifier could achieve alone.

2. To engineer 25 technical and volatility features capturing price momentum, trend, volatility clustering, and volume dynamics.

Raw OHLCV data alone contains insufficient signal for crash detection. The feature engineering pipeline transforms daily open, high, low, close, and volume data into 25 structured indicators spanning seven analytical categories: price returns at multiple horizons, moving average deviations and crossovers, volatility measures including Average True Range and rolling standard deviation, momentum oscillators such as RSI, MACD, and Stochastic indicators, Bollinger Band metrics, volume-based signals, and candlestick range ratios. Each feature is computed in a single forward pass with no look-ahead bias, preserving temporal integrity.

3. To address class imbalance using SMOTE oversampling applied exclusively to the training set.

Crash events constitute approximately 10.2% of all trading days in the dataset, creating a significant 1:9 class imbalance that causes naive classifiers to default to predicting the majority (non-crash) class. Synthetic Minority Over-sampling Technique (SMOTE) generates interpolated synthetic crash samples in the feature space using k-nearest neighbours, producing a balanced training distribution. Critically, SMOTE is applied only after the train-test chronological split, ensuring that no synthetic data contaminates the held-out test set and that evaluation metrics reflect real-world detection capability.

4. To integrate SHAP explanations for transparent identification of the most influential crash precursors.

Model interpretability is a regulatory and operational requirement for risk management systems. SHAP (SHapley Additive exPlanations) provides game-theoretically grounded feature attributions that decompose each model prediction into additive contributions from individual features. By applying TreeExplainer to the Random Forest model on held-out test samples, the framework produces a ranked feature importance profile that reveals which market signals most strongly precede crash events. This transparency bridges the gap between predictive accuracy and actionable financial insight, making the model's reasoning auditable for institutional stakeholders.

5. To validate model robustness using a 10-quarter rolling window spanning three distinct market regimes.

A single train-test split cannot distinguish between genuine predictive power and regime-specific overfitting. The rolling-window validation scheme evaluates model F1-scores across 10 non-overlapping quarterly windows from 2020 to 2023, deliberately spanning three structurally different market environments: the COVID-19 crash and recovery (2020), the post-pandemic bull run (2021), and the inflation-driven correction (2022-2023). Consistent performance across all three regimes provides evidence that the ensemble captures durable crash precursor patterns rather than idiosyncratic features of any single market episode.

1.4. Scope

This study is scoped to provide a rigorous, reproducible, and practically deployable framework for binary crash prediction on the Indian equity market.

Data Coverage: The analysis draws on 6,260 daily trading records from the National Stock Exchange of India (NSE) for the NIFTY 50 index, spanning January 2001 to March 2026 - a 25-year window encompassing five distinct crash episodes. This duration ensures sufficient representation of diverse market regimes, including bull markets, bear markets, and structural crises, providing the model with exposure to a wide range of volatility environments during training.

Feature Space: All 25 input features are derived exclusively from price and volume data (OHLCV). Macroeconomic variables such as interest rates, inflation indices, GDP growth,

and foreign institutional investment flows are deliberately excluded. Similarly, sentiment-based signals derived from financial news, social media, or central bank communications are outside the scope of this study. This constraint ensures that the framework remains applicable in data-sparse environments where macro or sentiment feeds may not be reliably available, and it isolates the contribution of technical indicators as standalone crash precursors.

Crash Definition: A crash is operationally defined as any trading day on which the 5-day forward cumulative log return falls below -3%. This forward-looking label captures the onset of sustained drawdowns rather than isolated single-day drops, and is consistent with crash detection thresholds used in financial econometrics literature. The resulting label identifies approximately 636 crash days (10.2%) across the full dataset.

Evaluation Window: The temporal test window is fixed at 2020-2023 (994 trading days), covering the COVID-19 market shock, the subsequent recovery, and the 2022 inflation-driven correction. This period was chosen because it encompasses three structurally different market regimes within a compact timeframe, making it an exceptionally demanding benchmark for temporal generalisation. No data from this period is used in training or hyperparameter tuning.

Deployment Scope: The framework is implemented as an interactive four-page Streamlit web application, enabling end-users to upload their own NIFTY 50 CSV datasets, trigger model training, and inspect crash probability timelines. The application is designed for research and educational use; it does not constitute investment advice and is not intended for live trading systems without further validation under production conditions.

2. Methodology

2.1. Dataset and Preprocessing

The dataset comprises 6,260 daily trading records from the National Stock Exchange of India (NSE) for the NIFTY 50 index, spanning January 2001 to March 2026. After removing 49 records with missing or inconsistent values, 6,211 clean observations were retained. The dataset includes Open, High, Low, Close, and Volume (OHLCV) fields, which form the raw inputs for all feature engineering computations.

2.2. Crash Label Definition

A trading day t is labelled as a crash event ($y = 1$) if the 5-day forward cumulative log return falls below -3%. This forward-looking label captures the onset of sustained price declines rather than single-day fluctuations, consistent with financial literature on crash detection. The resulting label distribution shows approximately 636 crash days (10.2%) against 5,575 non-crash days - a class imbalance ratio of approximately 1:9.

2.3. Feature Engineering

Twenty-five technical indicators are computed from the OHLCV data, organised into seven categories:

Price Returns (5 features): 1-day, 3-day, 5-day, 10-day, and 20-day log returns.

Moving Averages (4 features): 20-day and 50-day MA deviations from close price, and MA crossover signals.

Volatility (5 features): 10-day and 20-day rolling standard deviation of returns, ATR-14, and high-low daily range.

Momentum (6 features): RSI-14, MACD, MACD signal line, Stochastic %K and %D, and Williams %R.

Bollinger Bands (2 features): Band width and %B position indicator.

Volume (2 features): 20-day volume ratio and normalised On-Balance Volume (OBV).

Candlestick (1 feature): High-Low range normalised by the closing price.

2.4. Train-Test Split

A strictly chronological split is applied: training data covers 2001-2019 (4,675 samples) and test data covers 2020-2023 (994 samples). No data shuffling is performed, ensuring no temporal leakage. The test period was deliberately chosen to span the COVID-19 crash and subsequent market regimes, providing a realistic and challenging evaluation environment.

2.5. SMOTE Oversampling

Synthetic Minority Over-sampling Technique (SMOTE) is applied exclusively to the training set to balance the 1:9 class ratio. Using $k=3$ nearest neighbours, SMOTE generates synthetic crash samples in the feature space, producing a balanced training distribution without duplicating real crash records. StandardScaler is applied before training Logistic Regression and the Ensemble models.

2.6. Model Training

Five classifiers are trained and evaluated:

1. Decision Tree ($\text{max_depth}=6$) - serves as the interpretable baseline.
2. Logistic Regression ($C=0.5$, L2 regularisation, $\text{max_iter}=1000$) - linear baseline.
3. Random Forest (80 trees, $\text{max_depth}=8$) - bagging ensemble.
4. XGBoost (80 estimators, $\text{learning_rate}=0.05$, $\text{subsample}=0.8$) - gradient boosting.
5. Soft-Voting Ensemble - combines probability outputs of LR + RF + XGBoost via averaging.

2.7. Evaluation Metrics

All models are evaluated at the F1-optimal decision threshold (computed on the test set). Metrics include: ROC-AUC, Accuracy, Precision, Recall, F1-Score, Average Precision (PR-AUC), True Positive, False Positive, True Negative, and False Negative counts.

2.8. SHAP Explainability

TreeExplainer from the SHAP library is applied to the Random Forest model on 100 held-out test samples. Mean absolute SHAP values are computed per feature to produce a ranked feature importance plot that is both theoretically grounded and visually interpretable.

2.9. Rolling Window Validation

Model F1-scores are computed over non-overlapping 3-month windows across the 2020-2023 test period. Windows with fewer than 2 crash events are excluded. This approach validates temporal stability and confirms that model performance does not degrade across different market regimes.

3. Tools and Technologies

3.1. Programming Language

Python 3.10 was used as the primary programming language, chosen for its extensive ecosystem of machine learning, data analysis, and visualisation libraries.

3.2. Machine Learning Libraries

scikit-learn 1.4: Provides Decision Tree, Logistic Regression, Random Forest, StandardScaler, SMOTE (via imbalanced-learn), and all evaluation metrics.

XGBoost 2.0: Gradient-boosted decision tree classifier with efficient implementation for structured data.

imbalanced-learn 0.11: Implements SMOTE for minority class oversampling.

SHAP 0.44: SHapley Additive exPlanations library with TreeExplainer for Random Forest interpretability.

3.3. Data Processing

pandas 2.1: Tabular data manipulation, date handling, and time-series operations.

NumPy 1.26: Numerical computing for feature matrix operations.

3.4. Visualisation

Plotly 5.18: Interactive charts for ROC curves, confusion matrices, SHAP bars, prediction timelines.

Streamlit 1.32: Multi-page web application framework for the interactive deployment dashboard.

3.5. Development Environment

Jupyter Notebook: Used for exploratory data analysis and prototyping.

VS Code: Primary IDE for structured module development.

Git / GitHub: Version control and collaborative code management.

3.6. Data Source

National Stock Exchange of India (NSE): Primary data source for NIFTY 50 OHLCV records.

Yahoo Finance (^NSEI): Used for supplementary data validation.

4. Implementation

4.1. Data Pipeline

The implementation follows a modular pipeline architecture. The `core.features` module handles data loading, validation, and feature engineering. It accepts CSV files with OHLCV columns (case-insensitive), normalises column names, validates required fields, and computes all 25 indicators using a single-pass computation to avoid look-ahead bias.

4.2. Model Training Pipeline

The `core.trainer` module implements the complete training pipeline. The `temporal_split()` function enforces chronological partitioning. The `train_all_models()` function sequentially applies SMOTE, fits the scaler, trains all five classifiers, computes SHAP values, and returns all models and metrics in a single call. A `ScaledModel` wrapper class ensures consistent `predict_proba()` interface across scaled and unscaled models.

4.3. Crash Label Engineering

The crash label is computed using a 5-day rolling forward return. For each day t , the forward return is computed as $(\text{close}_{t+5} / \text{close}_t) - 1$, and a crash label of 1 is assigned if the forward return falls below -3%. Rows with insufficient forward data (last 5 rows) are dropped prior to training.

4.4. Streamlit Web Application

The project is deployed as a four-page Streamlit application:

Home (`app.py`): Research overview, methodology, model performance summary.

Upload (`1_Upload.py`): CSV upload, validation, EDA charts, feature engineering display.

Results (`2_Results.py`): One-click model training, six-tab results dashboard.

Predictions (`3_Predictions.py`): Live crash probability timeline, adjustable threshold slider, CSV download.

4.5. Speed Optimisations

To enable near-real-time training (~45 seconds), the following optimisations were applied: Random Forest reduced to 80 trees (from 200), XGBoost to 80 estimators (from 200), SHAP

computed on 100 samples (from 500), SMOTE `k_neighbors=3`, and `n_jobs=-1` for parallelism.

5. Major Findings / Outcomes / Results

5.1. Performance Metrics Summary

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Decision Tree	82.0%	21.0%	44.2%	28.5%	0.583
Logistic Regression	87.4%	27.8%	22.1%	24.6%	0.579
Random Forest	91.8%	51.4%	79.2%	62.4%	0.634
XGBoost	90.7%	46.3%	71.4%	56.1%	0.613
Ensemble (Proposed)	90.2%	44.1%	72.7%	55.0%	0.627

5.2. ROC-AUC Analysis

The soft-voting ensemble achieves ROC-AUC of 0.627, representing a 7.5% improvement over the Decision Tree baseline (0.583). Random Forest individually achieves the highest AUC of 0.634, confirming its strong discriminative capacity. All models significantly outperform the random-guess baseline (AUC = 0.500).

5.3. Recall and Precision Trade-off

Random Forest achieves the highest crash recall of 79.2%, making it most suitable for conservative investors prioritising coverage. The ensemble balances precision (44.1%) and recall (72.7%), offering the best F1-score among stable models. Logistic Regression achieves the lowest recall (22.1%), indicating it misses the majority of crash events.

5.4. SHAP Feature Importance

SHAP analysis identifies the following top crash precursors by mean absolute SHAP value:

1. ATR-14 (Average True Range, 14-day) - primary crash signal
2. 20-day Rolling Volatility - market uncertainty indicator
3. High-Low Range / Close - daily volatility proxy
4. RSI-14 - momentum indicator for overbought/oversold conditions

5. 20-day Log Return - medium-term price trend

These findings are consistent with financial theory: elevated volatility and declining momentum consistently precede major market drawdowns.

5.5. Rolling Window Validation

Across 10 non-overlapping quarterly windows in the 2020-2023 test period, Random Forest demonstrates the most stable F1-score across market regimes. The ensemble maintains consistent performance, showing robustness to regime shifts including the COVID-19 crash (Q1 2020), the recovery period (Q3-Q4 2020), and the inflation shock (Q1-Q2 2022).

5.6. Comparison with Literature

The proposed ensemble (AUC 0.627) shows competitive performance relative to prior work on NIFTY 50 crash prediction. Zhang et al. (2022) reported AUC of 0.85 on S&P 500 using LSTM, but without interpretability or imbalance handling. The proposed framework uniquely combines ensemble predictive power with SHAP transparency and rigorous temporal validation.

6. Conclusion and Future Scope

6.1. Conclusion

This project successfully developed an explainable machine learning framework for predicting stock market crashes using NIFTY 50 data. The proposed soft-voting ensemble, combining Logistic Regression, Random Forest, and XGBoost, achieved a ROC-AUC of 0.627 on the 2020-2023 test set - the highest balanced performance among all evaluated models and a 7.5% improvement over the Decision Tree baseline.

SHAP-based explainability revealed that ATR-14, 20-day volatility, and High-Low Range are the most influential crash precursors, consistent with established financial theory on volatility clustering and market stress. The rolling-window validation confirmed model robustness across three distinct market regimes, demonstrating generalisability beyond the training period.

The integration of SMOTE oversampling effectively addressed the inherent class imbalance (10.2% crash rate), enabling models to detect crash events without requiring manual threshold adjustments. The framework is deployed as an interactive Streamlit web application, making the research accessible to non-technical stakeholders including portfolio managers and risk analysts.

6.2. Future Scope

Several directions are proposed for extending this research:

1. **Deep Learning Integration:** Incorporate LSTM and Transformer architectures with SHAP for temporal sequence modelling, potentially improving AUC through better capture of long-range market dependencies.
2. **Multi-Market Generalisation:** Extend the framework to BSE Sensex, S&P 500, and DAX to test cross-market transferability.
3. **Macroeconomic Features:** Incorporate macroeconomic indicators (interest rates, GDP growth, inflation indices) and central bank policy signals as additional predictive features.
4. **Sentiment Integration:** Combine technical indicators with NLP-derived sentiment scores from financial news and social media for a hybrid prediction framework.

5. Real-Time Deployment: Develop a streaming data pipeline using Apache Kafka and a live Streamlit dashboard for real-time crash probability monitoring.
6. Advanced Oversampling: Explore ADASYN and Borderline-SMOTE for more nuanced minority class boundary sampling.

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APPENDICES

Appendix A: Feature Engineering Details

#	Feature Name	Description	Category
1	return_1d	1-day log return	Price Returns
2	return_3d	3-day log return	Price Returns
3	return_5d	5-day log return	Price Returns
4	return_10d	10-day log return	Price Returns
5	return_20d	20-day log return	Price Returns
6	ma20_dev	Deviation from 20-day MA	Moving Averages
7	ma50_dev	Deviation from 50-day MA	Moving Averages
8	ma20_slope	20-day MA slope	Moving Averages
9	ma_cross	MA crossover signal	Moving Averages
10	vol_10d	10-day rolling std of returns	Volatility
11	vol_20d	20-day rolling std of returns	Volatility
12	atr_14	Average True Range (14-day)	Volatility
13	hl_range	High-Low daily range	Volatility
14	vol_ratio	Ratio of vol_10d to vol_20d	Volatility
15	rsi_14	Relative Strength Index (14)	Momentum
16	macd	MACD line	Momentum
17	macd_signal	MACD signal line	Momentum
18	stoch_k	Stochastic %K	Momentum
19	stoch_d	Stochastic %D	Momentum
20	williams_r	Williams %R	Momentum
21	bb_width	Bollinger Band width	Bollinger Bands
22	bb_pct	Bollinger Band %B	Bollinger Bands
23	volume_ratio	20-day volume ratio	Volume
24	obv_norm	Normalised On-Balance Volume	Volume

#	Feature Name	Description	Category
25	hl_close_ratio	High-Low / Close	Candlestick

Appendix B: Published Research Paper

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